MIL-C-13481B
18 August 1983
SUPERSEDING
MIL-C-13481A
23 March 1966

MILITARY SPECIFICATION

CLAMPS, BEAM, RIGGER'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of rigger's beam clamp.

1.2 Classification. Clamps shall be the following sizes, as specified (see 6.2):

Size 1 - 10,000 pounds, safe working load
Size 2 - 20,000 pounds, safe working load

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 675 - Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The clamp shall be as shown in figure 1 and as specified herein. The throat of the clamp shall be formed by hot-bending.

3.2 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection (see 4.3 and 6.3).

3.3 Material. Components shall be fabricated from hot rolled carbon steel conforming to ASTM A 675, grade 70. Material shall be new and unused.

3.3.1 Material deterioration and control. The clamp shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Capacities. The clamp shall withstand a proof load equal to twice the following safe working loads without permanent deformation, cracks, permanent set, or splits in any component.

Size 1 - 10,000 pounds.
Size 2 - 20,000 pounds.

3.5 Finish. Unless otherwise specified (see 6.2), clamps shall have a commercial hot-rolled finish.

3.6 Identification marking. Clamps shall be legibly die-stamped with the manufacturer's name or tradename, "US", and the safe working load.

3.7 Workmanship. All components of the clamps including forgings, stampings, machined surfaces and welded parts shall be clean and free from dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered. The surfaces of all parts to be welded shall be free from scale, grease, and other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article clamp shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article clamp shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each clamp shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

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4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The clamps shall be examined as specified herein for the following defects:

Critical

1. Splits or cracks in any part of clamp.

Major

101. Materials not as specified or the materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
102. Contractor does not have documentation available for identification of material, material finishes or treatments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Components missing or not as specified.
105. Nonconformance to dimensions shown in figure 1.
106. Workmanship not as specified.

Minor

201. Finish not as specified.
202. Identification marking missing, incomplete, or illegible.

4.5.2 Proof test. Fasten the clamp to a channel beam or in a testing machine. Apply the proof load specified in 3.4 to the shackle hole. Nonconformance to 3.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Preservative not applied as specified for level A.
108.	Containers not as specified for level A or B.
109.	Clamps of unlike size packed together for level A or B.
110.	Clamps not immobilized within the container for level A or B.
111.	Strapping not as specified for level A.
112.	Preservation and packing not in accordance with the referenced document as specified for commercial.
113.	Marking missing, illegible, incorrect, or incomplete for level A, B or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Threaded surfaces of the bolt and any other unpainted surfaces of the clamp shall be coated with type P-1 preservative conforming to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.2 Commercial. The clamps shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The clamps of like size, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional; or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. The clamps shall be immobilized within the boxes by blocking, bracing, and anchoring in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless as otherwise specified strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The clamps shall be packed as specified in 5.2.1, except the boxes shall be domestic type or class as applicable. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

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5.2.3 Commercial. The clamps shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Clamps are intended for use in cargo handling operations to drag heavy, bulky items, large boxes, and loaded pallet boards into place in a ship's hold by block and tackle.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size of clamp required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Finish, if other than as specified (see 3.5).

e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

f. When other than finish B strapping is required (see 5.2.1).

6.3 First article. When a first article inspection is required, the item shall be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

Custodians:
Army - ME

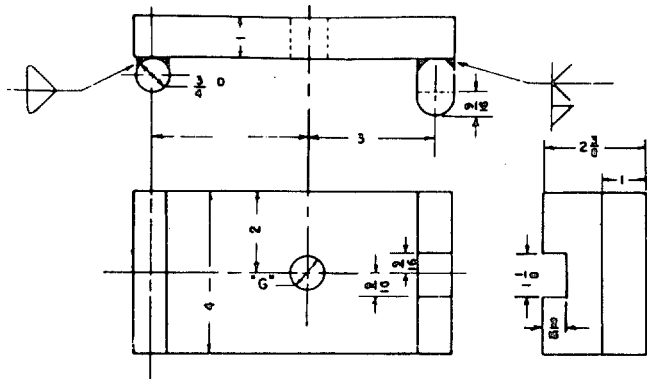
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Project 3940-0165

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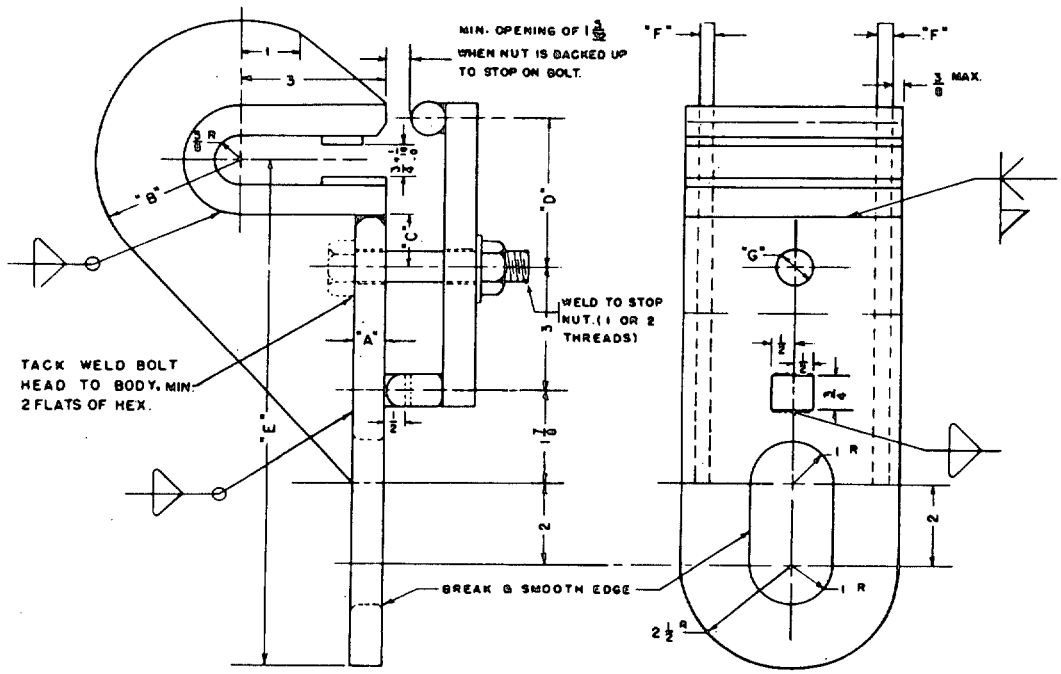


DOG DETAIL

SIZE	A	B	C	D	E	F	G	H
1	3/4	3 3/8	1 1/4	3 5/8	12	3/8	7/8	3/4
2	1	4 5/8	1 1/2	4 1/8	12 1/2	1/2	1	1

SIZE	BOLT SIZE	WASHER
1	3/4-10 NC2 X 5 LG.	3/4 PLAIN SAE MED.
2	7/8-9 NC2 X 5 LG.	7/8 PLAIN SAE MED.

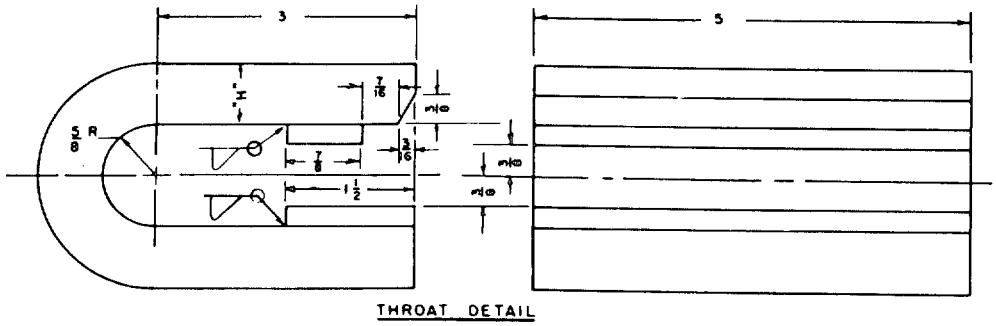
MIN. EFFECTIVE THREAD LENGTH OF 2 1/2 ON 3/4 DIA BOLT; 2 1/2 ON 7/8 DIA. BOLT.



CLAMP BODY

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE NOTED ALL TOLERANCES ARE ± 1/32.



THROAT DETAIL

FIGURE 1. Riggers beam clamp.

X-803A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-13481B	Clamps, Beam, Rigger's

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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